

# Smit Chaudhary

SOFTWARE ENGINEER · QUANTUM ALGORITHMS DEVELOPER · QUANTUM PHYSICIST

Rotterdam The Netherlands

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## Experience

### Quantum Algorithms Developer

Amsterdam, The Netherlands

PASQAL

2022 - Present

- Developed Quantum Scientific Machine Learning (QSciML) algorithms. Built and trained variational quantum circuits to solve industrially relevant aerodynamics PDEs, creating quantum alternatives to fully classical physics-informed neural networks (PINNs)
- Served as technical lead on European Union funded projects, driving collaboration with major aerospace industry partners
- Architected quantum software infrastructure and built hardware-aware circuit capabilities, making strategic technical decisions that aligned algorithms with neutral atom hardware constraints while enhancing both open-source and proprietary frameworks
- Mentored research interns and junior developers, leading to successful project contributions and published research outcomes

### Quantum Computing Intern

California, USA

MENTEN AI

May 2022 - Sept 2022

- Developed a novel fully quantum generative adversarial network (QGAN) architecture for discrete and binary data generation
- Investigated and implemented quantum circuit features such as noise reuploading in the generator and auxiliary qubits in the discriminator to enhance expressivity
- Demonstrated the model's performance on synthetic datasets and low-energy states of Ising models, showing successful data reproduction and generalization capabilities

## Education

### Technische Universiteit Delft

Delft, The Netherlands

MASTER OF SCIENCE IN APPLIED PHYSICS

August 2020 - July 2022

- **Thesis:** Quantum Walk Based Algorithms for Qubit Mapping (**Supervisor:** Prof. Sebastian Feld)
- **Research Experience:** Honors Project on Noise Induced Barren Plateaus at Leiden University (**Supervisor:** Prof. Jordi Tura)
- **Relevant Coursework:** Applied Quantum Algorithms, Quantum Information, Quantum Computing Architecture

### Indian Institute of Technology, Kanpur

Kanpur, India

BACHELOR OF SCIENCE IN PHYSICS

July 2016 - May 2020

- **Undergraduate Project:** Bohmian Mechanics and Quantum Information (**Supervisor:** Prof. Kaushik Bhattacharya)
- **Research Experience:** Summer Research at IISER Kolkata on Quantum Machine Learning (**Supervisor:** Prof. Prasanta Panigrahi)
- **Relevant Coursework:** Quantum Computing, Quantum Field Theory, Statistical Mechanics, Probability and Statistics, Optics

## Selected Publications

### Solving Fluid Dynamics Equations with Differentiable Quantum Circuits

S. Chaudhary, G. T. Balducci, O. Kyriienko, P. K. Barkoutsos, L. Cardarelli, A. A. Gentile

*Proceedings of the 35th Parallel CFD International Conference 2024*, May 2025

### Quantum Circuit Training with Growth-Based Architectures

C. Duffy, S. Chaudhary, G. V. Velikova

*arXiv preprint*, Nov 2024

### Towards a scalable discrete quantum generative adversarial neural network

S. Chaudhary, P. Huembeli, I. MacCormack, T. L. Patti, J. Kossaifi, A. Galda

*Quantum Science and Technology*, April 2023

## Skills

### Programming

Python, C/C++, Rust, Verilog

### Languages

English, Hindi, Gujarati

### Libraries & Utilities

Git, PyTorch, PennyLane, Tensorflow, Jax, Cirq, Qiskit